

Hassan Fawaz

5G & Telecom Systems Architect

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Professional Profile

Telecommunications systems architect and Ph.D. with experience in 5G Core/RAN integration, 5G NTN ground-segment engineering, requirements management, CONOPS, architecture documentation, design reviews, supplier selection, and end-to-end validation. Hands-on background in 3GPP-based 5G SA systems, roaming and inter-core connectivity, IP networking, cloud and container platforms, observability, cybersecurity, and AI/ML for network optimisation. Effective in international, cross-functional environments, translating operational and stakeholder needs into clear technical specifications, interface definitions, and executable test plans.

Core Expertise

Systems architecture: stakeholder needs; system/subsystem requirements; architecture baselines; CONOPS; interface definition; PDR/CDR; AIV and E2E validation; traceability; technical trade-offs; RFQ/TEB; supplier and partner coordination.

Telecommunications: 5G SA/NTN; 5G Core; RAN, gNB and UE; N32 roaming; 5GC-gNB and 5GC-5GC inter-connection; handover; IP/VPN; SD-WAN; cellular and full-duplex wireless networks; LoRaWAN.

Cloud, observability, and software: Linux; Docker; Kubernetes; Open5GS; Amarisoft; PacketRusher; ns-3/NR; OVHcloud; NETCONF; Prometheus; Grafana; Telegraf; Jira/Confluence; Git/Bitbucket; Python; C++; C; MATLAB; working knowledge of Java, JavaScript, HTML, and SQL.

Professional Experience

Univerty

2024–Present

5G Software Architect – France

- Translate mission, product, and partner needs into 5G NTN system and ground-segment requirements, architecture baselines, interface descriptions, performance specifications, CONOPS, and AIV/E2E validation plans in addition to supporting project and system PDR and CDR reviews.
- Manage a ground segment requirement database covering user terminals, gateways, gNB, 5G Core, and control-center elements; maintain specifications and create and manage requirement traceability.
- Coordinate the technical 5G Core supplier-selection cycle: requirements capture, market analysis, supplier workshops, RFQ preparation, offer analysis, Tender Evaluation Board contribution, final selection, and preparation of technical and contractual negotiations.
- Design and run interoperability and roaming experiments, including N9/N32 roaming and 5GC-gNB/5GC-5GC connectivity, using Open5GS, Amarisoft, PacketRusher, COTS, VPNs, Linux virtual machines, and cloud infrastructure.
- Define RAN and system procedures for handover, roaming, inter-operator operation, and RAN-sharing scenarios; present technical work in customer and partner progress reviews and support cross-functional architecture decisions.
- Defined security risks as well as baseline specification for the uniShape demonstrator mission including all systems and subsystems of the space, ground, and control segments.
- Assess system and cybersecurity risks and mitigations; contribute to business continuity, backup validation, cloud governance, IT continuity, and corporate AI policy.
- Worked on NETCONF-based telemetry and observability proof of concept with Telegraf, Prometheus, and Grafana,
- Worked on developing features for the digital twin of the constellation, a high-level emulator of the final system uniSky.

Telecom SudParis – Institut Mines-Telecom / Institut Polytechnique de Paris

2020–2024

Research Engineer and Lecturer – Palaiseau, France

- Designed and implemented deep reinforcement learning and graph-convolutional multi-agent methods for smart queue management and load balancing, helping maintain service-level agreements across classified network flows; results were published in IEEE TNSM and IFIP Networking.
- Worked on extending ns-3/NR towards 5G Standalone emulation through integration with Open5GS, alongside autonomous-network experimentation in academic and industrial collaborations.
- Delivered engineering and master's-level teaching on virtual networks, SD-WAN, Docker, and Kubernetes, including practical laboratory work.

Universite Paris-Saclay – UVSQ, Li-PaRAD

2019–2020

Postdoctoral Researcher – Versailles, France

- Developed multi-operator LoRaWAN channel and spreading-factor allocation methods using game theory, Nash and correlated equilibria, and LSTM-based time-series prediction.
- Published the resulting research in the IEEE Internet of Things Journal and Computer Networks.

Saint Joseph University, ESIB
2016–2019

Ph.D. Researcher and Teaching Assistant – Beirut, Lebanon

- Researched scheduling and power allocation in full-duplex OFDMA wireless networks, addressing fairness, spectral efficiency, imperfect channel-state information, and multi-cell scenarios through optimization, game theory, and reinforcement learning.
- Laboratory taught and supervised in C++, routing and switching, Linux, analog electronics, and UML for engineering and computer-science students.

Education

Saint Joseph University	2019
<i>Ph.D. – Scheduling and Power Allocation in Full-Duplex Wireless Networks</i> – Beirut, Lebanon	
Saint Joseph University	2016
<i>M.Sc. in Telecommunications, Networks, and Security – High Distinction</i> – Beirut, Lebanon	
Lebanese University	2015
<i>Engineering Diploma in Electrical and Communication Engineering – Distinction</i> – Beirut, Lebanon	

Selected Publications

Selected from 16 peer-reviewed publications; full list available through ORCID and the personal website.

• H. Fawaz et al., “Graph Convolutional Reinforcement Learning for Collaborative Queuing Agents,” <i>IEEE Transactions on Network and Service Management</i> , 2023.
• H. Fawaz et al., “Queue-Aware Resource Allocation in Full-Duplex Multi-Cellular Wireless Networks,” <i>IEEE Journal on Selected Areas in Communications</i> , 2023.
• K. Khawam et al., “A Channel Selection Game for Multi-Operator LoRaWAN Deployments,” <i>Computer Networks</i> , 2022.
• H. Fawaz et al., “Cooperation for Spreading Factor Assignment in a Multi-Operator LoRaWAN Deployment,” <i>IEEE Internet of Things Journal</i> , 2020.
• K. Khawam et al., “Non-Cooperative Edge Server Selection Game for Federated Learning in IoT,” <i>IEEE/IFIP NOMS</i> , 2024.

Selected Technical and Research Highlights

- Authored research on AI-driven network optimisation, multi-agent reinforcement learning, wireless resource allocation, IoT, and edge-server selection.
- Applied integer and mixed-integer optimisation, CVX/MOSEK, game theory, LSTM, DQN, and graph-convolutional reinforcement learning to telecommunications problems.
- Experienced in technical writing and review across requirements, system specifications, performance documents, CONOPS, validation plans, research papers, and funding proposals.
- Worked with internal engineering teams and external telecom operators, infrastructure providers, cloud providers, equipment vendors, research laboratories, and institutional customers.

Certifications and Awards

- Lebanese National Council for Scientific Research (CNRS-L) Doctoral Fellowship, 2016.
- Cisco Networking Academy course certificates: CCNA 1 Network Fundamentals; CCNA 2 Routing and Switching; CCNA 3 Scaling Networks.

Languages

Arabic: Native	English: Fluent	French: B2+, professional working proficiency
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